SCIENCE 8 · TERM TEST 1

Key Facts to Memorise

Chapter by chapter, in the order of the textbook. Learn the key words first, then the facts under them.



Chapter 1 · Respiration

1.1 The human respiratory system

Respiration: a series of chemical reactions that happens inside every living cell

Respiratory system: the organs that take oxygen out of the air and get rid of carbon dioxide

Cartilage: strong rings around the trachea that keep it open

Bronchiole: a smaller tube that a bronchus divides into

Larynx (voicebox): wide part at the top of the trachea; contains the vocal cords

Aerobic respiration: the usual kind of respiration in our cells; it uses oxygen

Trachea (windpipe): tube from the mouth and nose towards the lungs

Bronchus (plural bronchi): one of the two branches of the trachea; one goes to each lung

Air sac: tiny structure at the end of a bronchiole, where gas exchange happens

Vocal cords: bands of muscle stretched across the larynx; they vibrate to make sound

PATH OF AIR INTO THE LUNGS

nose or mouth → **larynx** → **trachea** → **bronchus** → **bronchiole** → **air sac (alveolus)**

Air going out of the lungs follows the same path in reverse.

FACTS

- Respiration is one of the characteristics shared by all living things.
- Aerobic respiration uses oxygen. The cells produce carbon dioxide as a waste product.
- When you breathe, air goes into your lungs and some of its oxygen goes into your blood. The blood delivers oxygen to every cell for respiration, collects the waste carbon dioxide and takes it back to the lungs.
- Air gets into the body through the mouth or nose. Both connect to the trachea.
- The rings of cartilage keep the trachea open and stop it collapsing, so air can keep moving in and out. You can feel them on the front of your neck.
- The trachea branches into two bronchi. The bronchi also have cartilage to support them.
- Each bronchus divides into several smaller bronchioles, which carry air deeper into the lungs.
- Bronchioles end by branching into many tiny air sacs. Here oxygen goes into the blood and carbon dioxide comes out.

- The vocal cords are like guitar strings: when they vibrate they make a sound. Your larynx changes when you make higher or lower sounds.

1.2 Gas exchange

Alveoli: another name for air sacs (one air sac is an alveolus)

Gas exchange: oxygen going from the air into the blood, and carbon dioxide from the blood into the air

Haemoglobin: the substance in red blood cells that oxygen combines with

Expired air: the air you breathe out

Analogy: something that stands for something else in a model (holes in jelly = air sacs)

Capillaries: very tiny blood vessels wrapped around the air sacs

Diffusion: particles moving from where there are a lot of them to where there are fewer

Inspired air: the air you breathe in

Limewater: liquid that goes cloudy with carbon dioxide

FACTS

- Under a microscope the lungs are mostly holes: the air sacs.
- The air sac wall is one layer of very thin cells. The capillary wall is also a single layer of very thin cells, and the capillary is pressed tightly against the alveolus.
- There are only two very thin cells between the air and the blood.
- Blood arriving at an air sac has come from the heart, and before that from the body's organs. Their cells respire, using oxygen and making carbon dioxide, so this blood has only a small amount of oxygen and a lot of carbon dioxide.
- Air in the air sac came from outside the body, so it has a lot of oxygen and only a small amount of carbon dioxide.
- Oxygen diffuses from the air, through the thin cells, into the blood. There it dissolves, goes into the red blood cells and combines with haemoglobin.
- Carbon dioxide diffuses the other way: from the blood (a lot) into the air in the air sac (a little).
- Lots of very small air sacs are better than a few large ones. The agar jelly experiment shows this: 32 small holes and 8 large holes have the same total volume, and the dye represents oxygen.
- Carbon dioxide makes limewater go cloudy. Limewater goes cloudy first in the tube that expired air bubbles through, so expired air contains MORE carbon dioxide than inspired air.

1.3 Breathing

Breathing: the movements that move air into and out of the lungs

Relax: a muscle returns to its normal size

Contract: a muscle gets shorter

FACTS

- The pressure of a gas increases when the volume of its container is decreased.
- Breathing IN: the intercostal muscles between the ribs contract and pull the ribs upwards and outwards. The diaphragm muscles contract and pull the diaphragm downwards.
- These two movements make more space inside the chest cavity: volume increases, pressure inside the chest and lungs decreases, and air moves down the trachea into the lungs to fill the space.
- Breathing OUT: the intercostal muscles relax and the ribs drop down to their natural position. The diaphragm muscles relax and the diaphragm becomes its normal domed shape.
- This makes less space: volume decreases, pressure increases, and air is squeezed out of the lungs.
- Volume of air breathed out can be measured with an upside-down bottle of water in a bowl, a volume scale marked using a measuring cylinder, and bendy tubing. Repeat three times and calculate the mean.
- Syringe model: pulling the plunger out increases the volume, so the balloon inflates. Balloon = lungs, hole in the bung = trachea, plunger = diaphragm, syringe barrel = rib cage. Unlike real ribs, the barrel does not move.

1.4 Respiration

Glucose: a sugar made when the digestive system breaks down carbohydrates; an energy store

Mitochondria: tiny structures inside cells that release energy from glucose by aerobic respiration

WORD EQUATION FOR AEROBIC RESPIRATION

glucose + oxygen → carbon dioxide + water

FACTS

- We use energy to move around, to send electrical impulses along neurones, and to keep warm when it is cold.
- All our energy comes from food. Carbohydrates are especially good for giving energy.
- The digestive system breaks carbohydrates down to glucose. Glucose goes into the blood, and the blood delivers it to every cell.
- Energy must be changed from one type to another, or transferred, in order to do something. The energy in glucose is locked up inside it, so it must be released before cells can use it.
- Most cells have many mitochondria. Aerobic means the reaction uses oxygen from the air.
- Some of the energy in glucose is released in a very controlled way, just a little at a time, just enough for the cell's needs.
- Every time energy is transferred, some of it is changed to heat. So respiring cells are a little warmer than their surroundings.
- All living things respire, even seeds. Germinating seeds respire especially quickly. Soaking peas in water for about an hour starts germination.
- Respiration happens in every living cell. Breathing is the movement of air into and out of the lungs. They are NOT the same thing.

1.5 Blood

Blood plasma: the liquid part of blood; very pale yellow, mostly water

Oxyhaemoglobin: very bright red compound formed when oxygen combines with haemoglobin

Pathogens: bacteria and viruses that can cause illness when they get into the body

Red blood cells: cells full of haemoglobin that carry oxygen; no nucleus, no mitochondria

White blood cells: cells that always have a nucleus and defend us against pathogens

Antibodies: chemicals made by some white blood cells that kill pathogens

FACTS

- Every cell needs a good supply of glucose and oxygen, and the carbon dioxide and water it makes must be taken away. The blood does the delivery and removal. The heart pumps constantly to keep blood moving inside blood vessels.
- Blood looks red because it contains a lot of red blood cells floating in the plasma. Most cells in blood are red blood cells: at least 20 trillion in an adult, about five million in every 1 cm^3.
- Plasma transports the red and white blood cells. Glucose dissolved in plasma travels from the digestive system to every cell. Carbon dioxide from respiration dissolves in plasma and is carried to the lungs, where it diffuses out and is breathed out.
- Red blood cells are very unusual: no nucleus and no mitochondria. No nucleus makes more space for haemoglobin; no mitochondria stops them using up the oxygen themselves.
- In the lung capillaries oxygen diffuses into the blood and then into red blood cells, where it forms oxyhaemoglobin. At respiring cells, oxyhaemoglobin lets go of its oxygen. The blood then returns to the lungs for more.
- Red blood cells are much smaller than most cells, so they get through very tiny capillaries, close to the alveoli and to respiring cells.
- Some white blood cells change shape and push out ‘fingers’ of cytoplasm to capture a pathogen, then make chemicals that kill and digest it.
- Other white blood cells make antibodies (drawn as Y-shapes). A different kind of antibody is needed for each kind of pathogen. Antibodies stick to the pathogen and either kill it directly or glue many pathogens together so they cannot move and other white blood cells can kill them.
- White blood cells cannot destroy the rabies virus in time. Without treatment most people infected with rabies die, so the body needs outside help such as vaccines.

Chapter 2 · Properties of materials

2.1 Dissolving

Dissolving: a solid seeming to disappear as its particles spread out in a liquid

Solute: the substance that dissolves

Solvent: the substance the solute dissolves into

Solution: the mixture formed when a solute dissolves in a solvent

Transparent: you can see through it (it can still be coloured)

Opaque: you cannot see through it

Conserved: kept: no mass is lost when something dissolves

FACTS

- A solution is a mixture. Sugar dissolved in water is still there: its particles have spread out among the water particles.
- A sugar crystal is visible because it is made of lots of groups of vibrating particles packed tightly together.
- Water particles vibrate and slide past one another. They bump into the sugar particles, separate them, and mix with them. Single sugar particles are too small to be seen.
- All solutions are transparent. Copper sulfate solution is blue but still transparent. Milk is opaque, so it is not a solution.
- Dissolving needs two substances, a solute and a solvent. Melting needs only one.
- mass of solute + mass of solvent = mass of solution. This is true for any solution.

2.2 Solutions and solubility

Concentrated: a solution with a lot of solute particles dissolved

Dilute: a solution with fewer solute particles dissolved

Soluble: a solid that dissolves in a solvent such as water

Insoluble: a solid that will not dissolve in water (e.g. iron filings)

Saturated: a solution in which no more of the solid will dissolve

Solubility: how much of a solute will dissolve in a solvent

FACTS

- Sodium chloride (common salt) and sugar are soluble in water. Iron filings are insoluble.
- Some soluble substances are more soluble than others: 100 cm^3 of water dissolves a lot of sodium chloride but only a tiny amount of lead chloride.
- To compare solubility, measure how much of each solute dissolves in a known amount of the same solvent at the same temperature.
- Most solutes dissolve more quickly and easily in hot water. Particles with more energy vibrate and move more.

- As temperature increases, the solubility of most solutes increases: 100 g of water dissolves 204 g of sugar at 20 °C but 362 g at 80 °C.
- Water is not the only solvent. Oil paint brushes are cleaned with methanol (methylated spirits). Most nail polish remover contains propanone (acetone).
- When measuring a liquid, use a pipette to add or remove the final amount drop by drop for accuracy.

2.4 Paper chromatography

Paper chromatography: a technique that separates dissolved substances, such as the inks in black ink

Chromatogram: the image produced on the paper

Solvent front: the point that the solvent reaches on the paper

Permanent: ink (e.g. in marker pens) that is not soluble in water

FACTS

- Black ink is a mixture of different coloured inks.
- As water (the solvent) moves up the paper it dissolves the inks and carries their particles. Inks with different solubility are carried different distances. The more soluble the ink, the further its particles are carried.
- Method: pencil line about 1 cm from the end; small ink spot on the line (dry it, add a little more); about 2 cm of water in a beaker; hang the paper so the spot stays above the water; remove the paper before the water reaches the top (wet paper tears easily); let it dry.
- Inks that are not soluble in water (permanent markers) need a different solvent, such as alcohol. The green pigment in spinach is dissolved in ethanol. Use ethanol in a well-ventilated room with no heat sources nearby.
- A pure substance gives only one spot. A mixture separates into two or more spots.
- Scientists use chromatography to check food dyes, hair dyes and inks, because some substances could be toxic or cause allergic reactions. They compare the chromatogram with chromatograms of permitted colourings.
- Words that look alike: solute / solvent / solution; chromatography (the technique) / chromatogram (the result); dissolve / dilute.

WS2 Also tested in Worksheet 2 (not in the textbook pages)

Unsaturated: a solution that can still dissolve more solute

Alloy: a mixture of metals, e.g. brass = zinc dissolved in copper (a solid in a solid)

Residue: the solid left in the filter paper

Evaporation / crystallisation: heating a solution to remove the water and leave the solute as crystals

Aqueous solution: a solution in which water is the solvent

Filtration: separates an insoluble solid (e.g. sand) from a liquid

Filtrate: the liquid that passes through the filter paper

Rf value: distance moved by the spot ÷ distance moved by the solvent front; always less than 1

FACTS

- Dissolving is a physical change: no new substance is made, and the solute can be recovered by evaporation.
- A solid dissolves faster if the water is hotter, if it is stirred, or if the solid is crushed into smaller pieces. A larger lump dissolves more slowly.
- Hotter water particles have more energy and move faster, so they hit the solid more often and with more force.
- Solutions are not only solids in liquids: a fizzy drink is a gas (carbon dioxide) in a liquid, and brass is a solid (zinc) in a solid (copper).
- Most solids become MORE soluble as temperature rises, but gases become LESS soluble. That is why a warm fizzy drink goes flat.
- To test whether a solution is saturated, add a little more solute and stir. If it stays undissolved, the solution is saturated.
- To make a solution more concentrated without adding solute, evaporate some of the water. To make it more dilute, add more water.
- Sand, salt and water: filter to remove the sand (the residue), then evaporate the filtrate to get salt crystals.
- When a hot saturated solution cools, solute comes out as crystals. Mass of crystals = solubility at the high temperature – solubility at the low temperature (for 100 g of water).
- Potassium nitrate's solubility rises steeply with temperature; sodium chloride's hardly changes (about 36 g to 40 g from 0 °C to 100 °C).
- The solvent must start BELOW the baseline, otherwise the spots dissolve into the solvent in the beaker.

Not covered in these notes

Topic 2.3 (textbook pages 53–56) was not in the scanned pages. The Worksheet 2 section above covers what that worksheet tests (filtration, evaporation, crystallisation, types of solution), but other parts of Topic 2.3 may still be missing.

The textbook gives two different solubilities for copper sulfate at 20 °C: 32 g per 100 g water in the first table and 17 g in the temperature table. Ask the teacher which value to learn.